MAJOR PROJECT REPORT

On

“Resume IQ (AI Resume Analyzer and Job Recommendation)”

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DECLARATION

Certified that this project/training report entitled “Resume IQ (AI Resume Analyzer and Job Recommendation)” submitted by Khushpreet Kaur (2300733), student of B.Tech. CSE, Bhai Gurdas Institute of Engineering & Technology, Sangrur under I.K. Gujral Punjab Technical University, Jalandhar is an authentic record of my work as a part of B.Tech (CSE) 7th semester under the supervision of Mr. Maninder Singh, Training Incharge-cum-Software Developer at Solitaire Infosys Private Limited, Patiala (Punjab). This report has not been submitted to any other university or institution for the award of any degree.

(Signature of student)

Date:

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ABSTRACT

Resume IQ (AI Resume Analyzer and Job Recommendation) is a full-stack web application developed using the MERN stack (MongoDB, Express.js, React.js and Node.js). The platform helps students and job seekers understand how effectively their resumes present their qualifications and how well they match available job roles. A user uploads a resume in a supported format such as PDF, DOCX or TXT. The application processes the uploaded file, extracts relevant information such as education, skills, experience and projects, and evaluates the resume to generate an overall score out of 100.

The system identifies skills mentioned in the resume, highlights areas that may need improvement and provides suggestions for additional skills or sections that could strengthen the candidate profile. Based on the detected skills and experience, Resume IQ compares the candidate profile with relevant job roles and displays matching percentages along with direct application links where available. React.js provides the interactive user interface, while Node.js and Express.js handle server-side processing and API requests. MongoDB is used for storing application data where required. The application aims to make resume evaluation easier, provide useful career guidance and help users discover job opportunities that better match their profiles.

Keywords: Resume Analysis, MERN Stack, Resume Score, Skill Detection, Job Recommendation, Full-Stack Web Application.

TABLE OF CONTENTS

CONTENTS	PAGE
ABSTRACT	4
TABLE OF CONTENTS	5
1. ORGANIZATION PROFILE	6
2. INTRODUCTION	8
3. OBJECTIVES AND SCOPE	9
4. TECHNOLOGY USED	10
5. SYSTEM ARCHITECTURE AND WORKFLOW	11
6. DATABASE DESIGN	12
7. MAJOR FEATURES AND MODULES OF RESUME IQ	13
8. IMPLEMENTATION AND SECURITY	14
9. TESTING AND RESULTS	15
10. SCREENSHOTS AND RESULTS	17
11. LIMITATIONS AND FUTURE OUTCOMES	21
12. CONCLUSION	22
13. REFERENCES	23

1. ORGANIZATION PROFILE

1.1 An Overview

SOLITAIRE INFOSYS is a leading Software and Web Application Development Company, based in Mohali (Chandigarh) that provides high quality comprehensive services to enterprises across a wide range of platforms and technologies. Our major areas of expertise are in providing quality, cost effective software or web development.



Figure 1.1: Solitaire Infosys office building

1.2 Focus of the Company

Figure 1.2: Company logo

The Company focuses on understanding the diverse and mission-critical needs of each of our clients. To understand is to be able to deliver. The competence and experience of our company gives us a competitive edge by making sure we provide the best services and products to our clients. Our high quality standards enable us to deliver reliable and error-free software. The Company provides Web design/Web development, B2B & B2C E-commerce solutions, SEO & Web Promotions strategies implementation consulting services to both domestic as well as international clients at the most affordable rates.

1.3 Operating Excellence

- A safe, fun and professional work environment
- Good relationships with industry and other partners
- Respect for the rights and ambitions of our employees
- Honorable, value-driven business relations

1.4 History of the Company

Solitaire Infosys Inc. was a dream that came into existence over three years ago with a strong aspiration of becoming the best IT service provider around the globe. Presently, Solitaire Infosys is already leading the race with its competitors. Being nurtured by a team of experienced and sensitive people, the company tries to bond

emotionally with its clients and goes the extra mile to satisfy their needs. At Solitaire Infosys Inc. the team provides experience, expertise and capabilities that enable organizations to accelerate their service processes, deliver more service value and realize breakthrough results in the markets they serve. Clients appreciate its comprehensive range of services, innovative, streamlined and cost-effective solutions, and high level of customer service.

1.5 Project Development Environment

Resume IQ was developed in a software development environment using modern full-stack web technologies. The application allows a user to upload a resume in a supported file format. The website processes the resume, identifies important details such as skills, education and experience, and provides an overall resume score. It also compares the detected profile with suitable job roles and presents matching opportunities with application links. The project uses React.js for the frontend, Node.js and Express.js for backend services, and MongoDB for database management.

2. INTRODUCTION

2.1 Project Overview

Resume IQ is a responsive web application developed using the MERN stack to help users evaluate their resumes and discover suitable job opportunities. The user uploads a resume, and the system analyses its content to identify skills, education, experience and other relevant details. It then generates a resume score out of 100 and provides suggestions that can help improve the resume. Based on the skills and qualifications detected, the application matches the user with relevant job roles, displays match percentages and provides direct application links where available.

2.2 Introduction

A resume is an important document in the job application process because it presents a candidate's education, skills, experience and achievements. Many students and job seekers are unsure whether their resume is complete, whether it highlights the right skills or which job roles are suitable for their profile. Resume IQ provides a single, easy-to-use platform for resume evaluation and job discovery. It reduces the effort required to manually review a resume and search for matching roles by presenting an organized analysis and recommendations.

2.3 User Experience

Resume IQ is designed with a clean and simple interface. Users can navigate to the analysis section, upload a resume and start the analysis process. The result page presents the resume score, detected skills, improvement suggestions and job matches in an understandable format. The responsive interface is intended to provide a consistent experience on desktop, tablet and mobile screens.

2.4 Need for Resume Analysis and Job Recommendation

Job seekers often find it difficult to judge the quality of their resumes and identify roles that match their current skills. Manual checking can take time and may overlook missing sections or relevant keywords. Searching across multiple job platforms can also make it difficult to compare opportunities. Resume IQ addresses these challenges by analysing resume information, highlighting improvement areas and suggesting job categories based on the user profile. The final job selection and application decision remain with the user.

3. OBJECTIVES AND SCOPE

3.1 Objectives

- To provide a simple platform for uploading resumes in supported formats.
- To analyse resume content and identify skills, education, experience and key sections.
- To generate an understandable resume score out of 100.
- To provide suggestions that can improve resume quality and presentation.
- To match detected skills and qualifications with relevant job roles.
- To display job-match percentages and direct application links where available.
- To build a responsive and user-friendly full-stack web application using the MERN stack.

3.2 Scope of the Project

- The system accepts resume files in supported formats such as PDF, DOCX and TXT, subject to the application file-size and validation rules.
- Resume content is processed to identify important candidate information and skills.
- The application provides a score and improvement suggestions based on the information it can detect.
- The detected profile is compared with supported job roles to generate relevant matches.
- Job recommendations may include match percentages and links to external job application platforms.
- The project can be extended with more job sources, additional resume formats, multilingual analysis and personalized career guidance.

4. TECHNOLOGY USED

4.1 Technology Overview

Resume IQ is a full-stack web application developed using the MERN stack, which consists of MongoDB, Express.js, React.js and Node.js. These technologies work together to provide the user interface, server-side processing, API communication and database functionality.

4.2 Frontend Technologies

React.js: A JavaScript library used to build interactive and reusable user interface components. It helps display upload forms, analysis results, resume scores and job recommendations.

HTML: Used to define the structure and content of the web pages, including headings, forms, buttons and result sections.

CSS: Used to style the application, manage layouts, provide visual consistency and make the interface responsive across different screen sizes.

JavaScript: Adds functionality to the website, handles user interactions and supports communication with backend APIs.

4.3 Backend Technologies

Node.js: A JavaScript runtime used to execute server-side code and handle application requests.

Express.js: A lightweight Node.js framework used to create backend routes and REST APIs. It receives requests from the frontend, validates data and coordinates resume analysis and job matching operations.

4.4 Database Technology

MongoDB: A NoSQL document database used to store application information where required. Its flexible document structure is suitable for storing user-related data, analysis records and other application information.

4.5 Development and Version Control Tools

Git and GitHub can be used to track source-code changes, maintain project versions and host the project repository. A code editor such as Visual Studio Code supports writing, debugging and managing the application files.

5. SYSTEM ARCHITECTURE AND WORKFLOW

5.1 System Architecture

Resume IQ follows a client-server architecture based on the MERN stack. The React.js frontend provides the interface through which users upload resumes and view results. The frontend sends requests to the backend through REST APIs. The Node.js and Express.js backend validates requests, processes uploaded resume data, coordinates skill identification and job matching, and communicates with MongoDB when application data needs to be stored or retrieved. The processed results are returned to the frontend for display.

5.2 Application Workflow

1. User opens the Resume IQ website and navigates to the resume analysis section.
2. User selects and uploads a supported resume file.
3. The application validates the file type and size before processing it.
4. The backend extracts and analyses relevant resume information, including skills, education and experience.
5. The system calculates a resume score out of 100 and prepares improvement suggestions.
6. The detected skills and profile information are compared with supported job roles.
7. The result page displays the resume score, detected skills, suggestions, matching jobs and available application links.

5.3 Main Components

Component	Responsibility
React.js frontend	Resume upload interface, navigation and presentation of analysis results.
Node.js server	Runs backend JavaScript and handles application processing.
Express.js APIs	Manages routes, request validation and communication between frontend and backend.
Resume analysis module	Extracts relevant resume details and prepares score and improvement feedback.
Job matching module	Compares detected skills with job-role requirements and calculates match information.
MongoDB	Stores application records and other required structured data.

6. DATABASE DESIGN

6.1 Database Overview

Resume IQ uses MongoDB, a NoSQL database, for managing application data. MongoDB stores information in flexible document-based collections. The exact collections depend on the implemented application modules. A typical database structure for Resume IQ may include user information, resume analysis records and job-role information. The backend can interact with MongoDB through Mongoose models to validate and manage documents.

6.2 Suggested Collections and Their Purpose

Collection	Purpose
users	Stores registered user details and authentication-related information, if user accounts are enabled.
resumeAnalyses	Stores analysis records such as extracted skills, education, experience, score and improvement suggestions, if analysis history is enabled.
jobs	Stores or caches job-role details, required skills, descriptions and application URLs used for matching.

6.3 Data Flow

When a user uploads a resume, the frontend sends the file to the backend through an API request. The backend validates the file and processes its contents to identify relevant information. The analysis module prepares the resume score and improvement suggestions. The job matching module compares the extracted skills and profile details with available job-role requirements. If the application stores user or analysis information, the backend saves the relevant records in MongoDB. Finally, the backend returns the analysis and job recommendations to the frontend for display.

6.4 Data Validation and Privacy

The application should validate uploaded file types and file sizes and handle invalid or unreadable files with clear messages. Resume information may contain personal details, so it should be processed and stored only as required by the application. Access to stored records should be restricted to authorized users, and sensitive configuration values should be kept outside publicly accessible source code.

7. MAJOR FEATURES AND MODULES OF RESUME IQ

7.1 Resume Upload

The resume upload module allows users to select a resume file in a supported format such as PDF, DOCX or TXT. The interface provides a clear upload area and guides the user through the analysis process.

7.2 Resume Content Analysis

The analysis module processes the uploaded document and identifies relevant details such as contact information, education, skills, experience and projects. The quality of the extracted information depends on the file format and clarity of the resume.

7.3 Resume Score

Resume IQ presents an overall score out of 100 to give users a quick indication of how effectively their resume covers important information and sections. The score is intended as guidance and does not guarantee selection by an employer.

7.4 Skill Detection

The system identifies skills mentioned in the resume and organizes them so users can understand which technical or professional abilities are already represented in their profile.

7.5 Improvement Suggestions

The application provides suggestions for improving the resume, such as adding relevant skills, strengthening descriptions or completing important sections where information appears to be missing.

7.6 Job Matching and Recommendations

The job matching module compares the skills and profile information found in the resume with supported job roles. It displays suitable roles with match percentages to help users understand how closely their profile aligns with each role.

7.7 Direct Application Links

Where available, the result page provides links to external job platforms or application pages. Users can open these links to review job details and complete the application on the relevant platform.

7.8 Responsive User Interface

The website is designed to present the upload form and analysis results in a clear layout across common desktop and mobile screen sizes.

8. IMPLEMENTATION AND SECURITY

8.1 Frontend Implementation

The frontend is developed using React.js, HTML, CSS and JavaScript. It contains the navigation bar, landing section, resume upload interface, analysis action and result presentation components. The interface displays the resume score, detected skills, improvement suggestions and job matches in a structured manner. Reusable React components help keep the interface consistent and easier to maintain.

8.2 Backend Implementation

The backend is developed using Node.js and Express.js. It provides API endpoints to receive resume uploads, validate incoming requests, process resume information and return analysis results. It also coordinates the job matching process and communicates with MongoDB when data storage or retrieval is required.

8.3 Resume Processing and Job Matching

The uploaded file is checked before processing. The application extracts readable text and identifies relevant resume information. Based on the extracted content, the analysis logic prepares a score and improvement suggestions. Skills and other profile details are then compared with the requirements associated with available job roles. Matching results are sent to the frontend with the relevant role names, match information and application URLs where available.

8.4 Security and Error Handling

- Validate file extensions, MIME types and upload size on the server.
- Avoid exposing API keys, database credentials or other secrets in frontend code or public repositories.
- Use secure configuration and access controls for any stored user or resume data.
- Handle unsupported, damaged or unreadable files with understandable error messages.
- Use HTTPS when deploying the application and restrict access to protected backend routes where authentication is implemented.

9. TESTING AND RESULTS

9.1 Testing Overview

Functional testing is used to evaluate the major modules of Resume IQ. Testing focuses on the main workflow, including opening the website, uploading supported resume files, validating file formats, processing resume content, generating the resume score, identifying skills, displaying improvement suggestions and matching job roles. Testing also covers the presentation of application links and communication between the frontend, backend APIs and database. Any errors identified during testing should be corrected and retested to improve the reliability and consistency of the application.

9.2 Test Cases

Test Case	Test Operation	Expected Result
TC01	Open website	The home page loads and navigation elements are visible.
TC02	Upload PDF resume	A supported PDF file is accepted for analysis.
TC03	Upload DOCX or TXT resume	A supported document or text file is accepted.
TC04	Upload unsupported file	The application rejects the file and displays a clear validation message.
TC05	Upload empty or unreadable file	The application displays an appropriate error or asks the user to upload a valid resume.
TC06	Analyze valid resume	The system processes the resume and displays analysis results.
TC07	Generate resume score	A score out of 100 is displayed with the analysis result.
TC08	Detect skills	Skills found in the resume are displayed in an organized manner.
TC09	Show improvement tips	The application displays relevant suggestions based on the available resume content.
TC10	Match job roles	Relevant job roles and their match percentages are displayed.
TC11	Open application link	The selected job application link opens the corresponding external page when available.
TC12	Backend/database communication	Valid requests are processed and required data is retrieved or stored as configured.

9.3 Testing Result

The testing process is intended to verify that the major functions of Resume IQ work together correctly. The focus is on the resume upload and validation workflow, content analysis, score generation, skill detection, improvement suggestions, job matching and application links. Testing should be repeated after fixing any issues to confirm that the application provides consistent results for supported files. The accuracy of resume

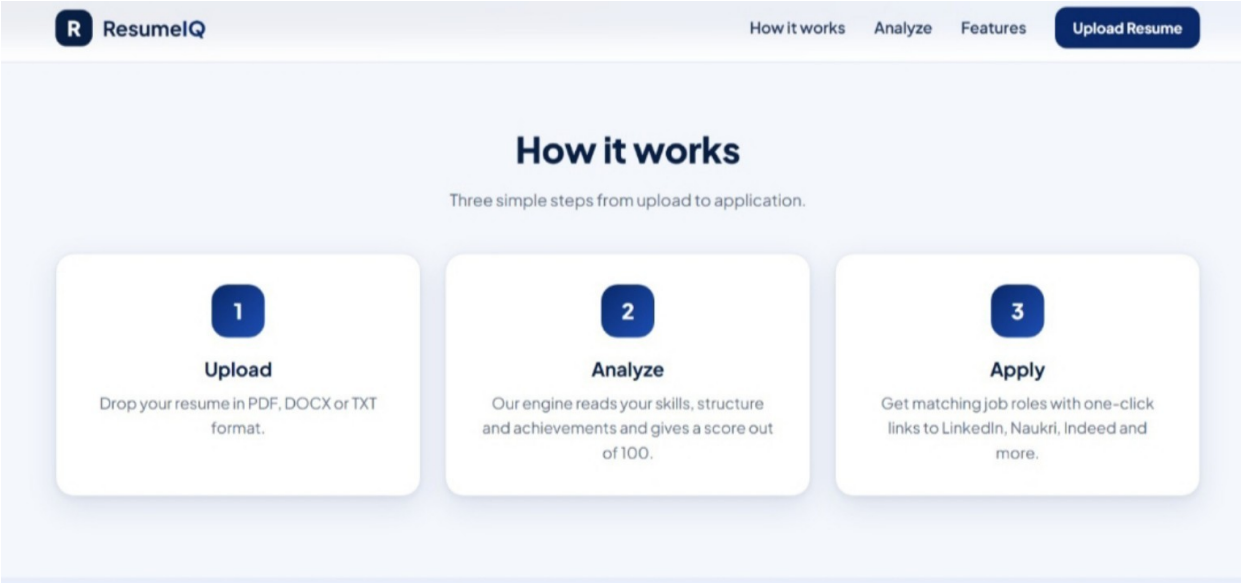
evaluation and job matching depends on the quality of extracted text and the job-role data available to the application.

9.4 Remaining Improvements

- Automated Testing: Introduce unit, integration and end-to-end tests for frontend components, backend APIs and file-processing operations.
- Enhanced Analysis Validation: Evaluate score consistency and improve the quality of extracted skills and recommendations.
- Security Improvements: Add strict file validation, secure upload handling, security headers and improved error handling.
- Performance Testing: Test larger files and multiple concurrent requests to evaluate response time and scalability.
- Job Data Updates: Keep job-role information and external application links current and validate links periodically.

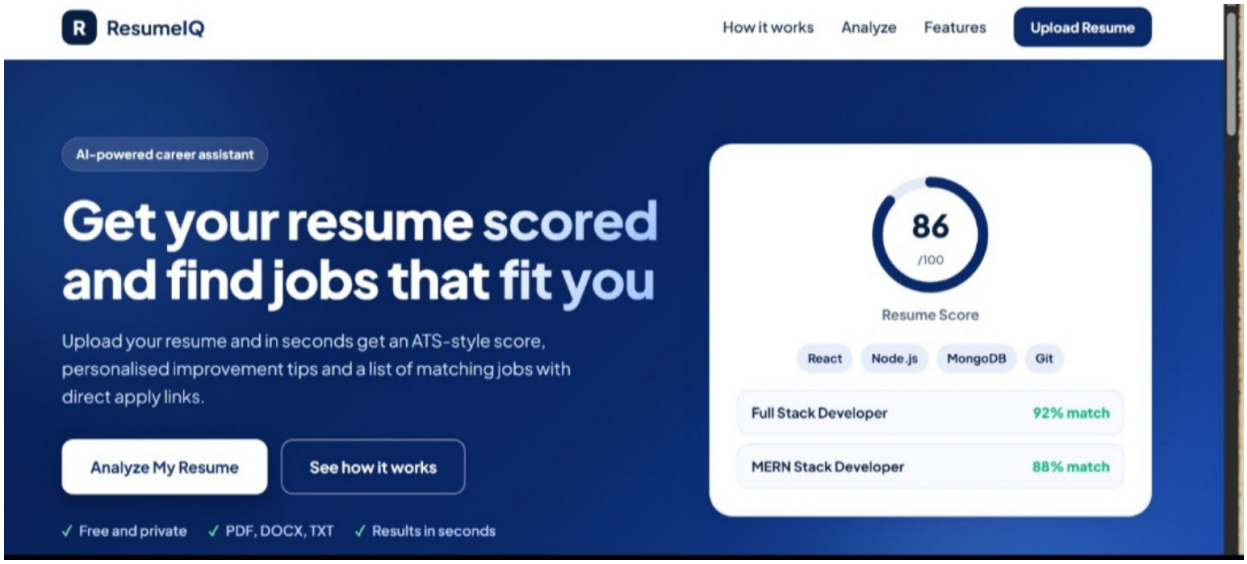
10. SCREENSHOTS AND RESULTS

Figure 10.1: Resume IQ Home Page



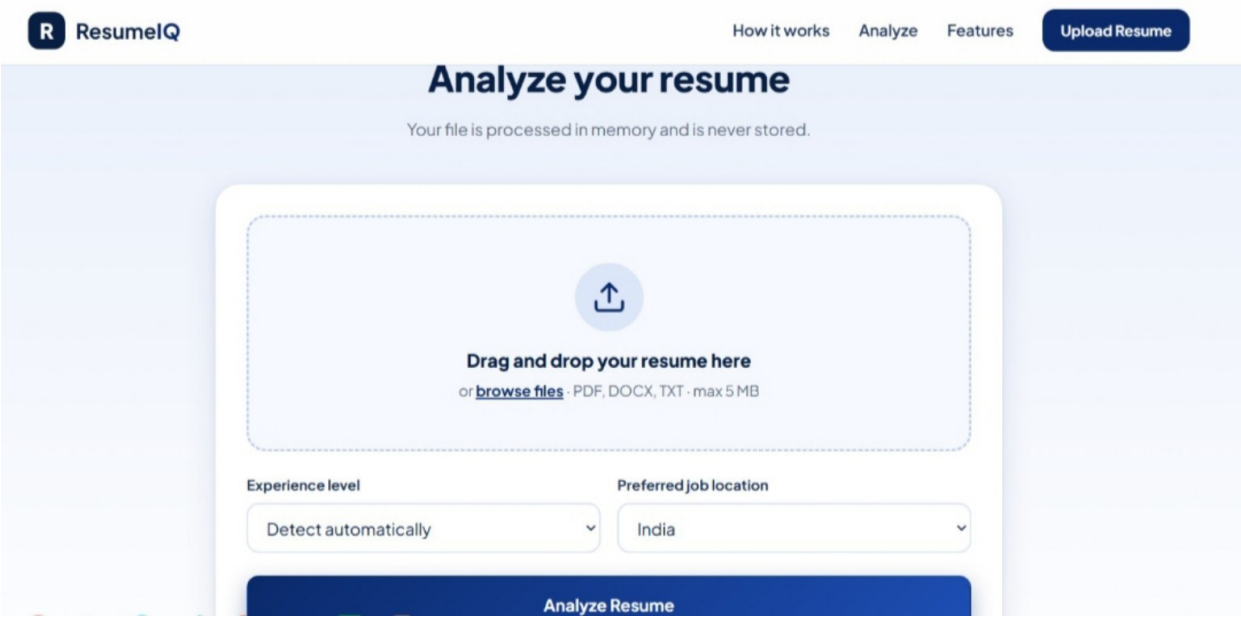
The home page introduces Resume IQ as an AI-powered career assistant. It explains that users can upload a resume to receive an ATS-style score, personalized improvement tips and matching job recommendations. The page also highlights the main navigation options and provides clear buttons to begin resume analysis or learn how the platform works.

Figure 10.2: Analyze Your Resume – Upload Section



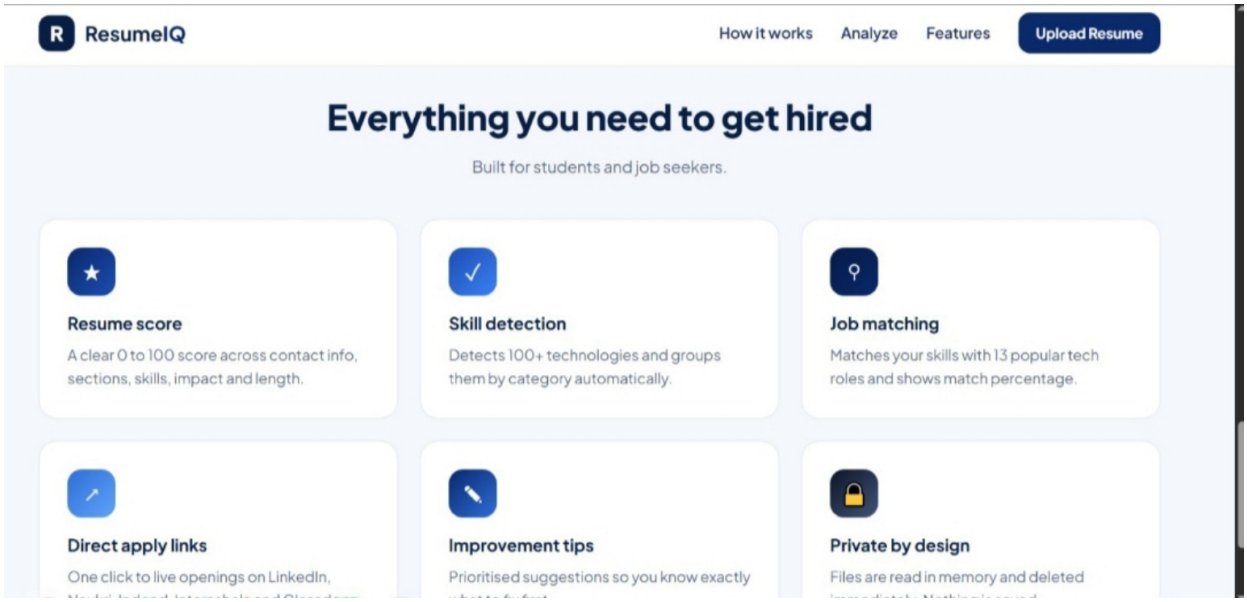
This screen provides the resume upload interface. Users can drag and drop a file or browse their device to select a supported PDF, DOCX or TXT resume. The page also allows users to select an experience level and preferred job location before starting the analysis.

Figure 10.3: How Resume IQ Works



This section explains the application workflow in three simple steps: upload the resume, analyse its skills and structure to generate a score out of 100, and view matching job roles with links to job platforms. The step-based layout helps users understand the process before using the analyzer.

Figure 10.4: Resume IQ Features



This section presents the main features of Resume IQ, including resume scoring, skill detection, job matching, direct application links, improvement tips and privacy-focused file handling. The feature cards summarize the benefits available to students and job seekers.

11. LIMITATIONS AND FUTURE OUTCOMES

Resume IQ provides a useful starting point for resume evaluation and job discovery. However, the quality of its output depends on the clarity and structure of the uploaded document, the accuracy of text extraction and the job information available to the matching module. A resume score or match percentage is guidance only and cannot predict an employer's final hiring decision.

11.1 Improved Resume Parsing

Support additional document formats and improve extraction from complex layouts, tables, columns and scanned documents through suitable text recognition methods.

11.2 More Detailed Feedback

Provide more specific suggestions for different industries, experience levels and job descriptions, including stronger achievement statements and keyword guidance.

11.3 Expanded Job Sources

Connect additional trusted job platforms or job data providers to increase the range and freshness of recommended opportunities.

11.4 Personalized Recommendations

Improve matching by considering experience level, education, preferred location and career interests in addition to technical skills.

11.5 Privacy and Security

Continue improving secure file handling, data minimization, access controls and clear user information about how resume data is processed.

11.6 Performance and Scalability

Optimize file processing and backend APIs to support larger resumes and a higher number of simultaneous users.

11.7 Overall Future Outcome

With these enhancements, Resume IQ can develop into a more comprehensive career-support platform that helps students and job seekers improve their resumes, understand their skill profiles and discover relevant employment opportunities.

12. CONCLUSION

In conclusion, Resume IQ (AI Resume Analyzer and Job Recommendation) is a full-stack web application developed to make resume evaluation and job discovery easier for students and job seekers. Built using the MERN stack (MongoDB, Express.js, React.js and Node.js), the platform provides a simple process in which a user uploads a resume and receives an analysis of its important information. The application identifies skills, presents a resume score out of 100 and provides suggestions that can help improve the document.

The job recommendation feature compares the skills and profile details detected in the resume with relevant job roles. It displays matching percentages and provides direct application links where available, helping users explore opportunities that align with their qualifications. The project combines frontend interface development, backend API processing, resume analysis, database management and job matching into one web application.

The project demonstrates how full-stack web technologies can be used to solve a practical career-related problem. Although resume scoring and job matching should be treated as guidance rather than a guarantee of employment, Resume IQ offers users a convenient starting point for improving their resumes and identifying suitable roles. Future improvements in parsing accuracy, job-source integration, security and personalization can make the platform more useful and reliable.

13. REFERENCES

1. React Documentation: https://react.dev
2. Node.js Documentation: https://nodejs.org/docs/latest/api/
3. Express.js Documentation: https://expressjs.com
4. MongoDB Documentation: https://www.mongodb.com/docs/
5. Mongoose Documentation: https://mongoosejs.com/docs/
6. JavaScript Documentation (MDN Web Docs): https://developer.mozilla.org/en-US/docs/Web/JavaScript
7. HTML Living Standard (WHATWG): https://html.spec.whatwg.org/
8. CSS Documentation (MDN Web Docs): https://developer.mozilla.org/en-US/docs/Web/CSS
9. Git Documentation: https://git-scm.com/doc
10. GitHub Documentation: https://docs.github.com/